

Coding & Solution

Date	4 July 2026
Team ID	xxxxxx
Project Name	OptiCrop – Smart Agricultural Production Engine project.
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/hema-chitra/OPTICROP-SMART-AGRICULTURAL-PRODUCTION-OPTIMIZATION-ENGINE-ALML
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn

Key Features Implemented	• Crop recommendation using a Random Forest Machine Learning model. • User-friendly web interface for entering soil and weather parameters. • Instant crop prediction based on user inputs. • Input validation for accurate predictions. • Responsive web design using HTML and CSS.
Pending / Incomplete Features	• User authentication and login. • Real-time weather API integration. • Fertilizer recommendation system. • Mobile application and cloud deployment.
Setup / Run Instructions	1. Install Python and required libraries using pip install -r requirements.txt. 2. Open the project folder in Visual Studio Code. 3. Run the application using python app.py. 4. Open http://127.0.0.1:5000 in a web browser. 5. Enter soil and weather values, then click Predict Crop to view the recommended crop.



Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The **OptiCrop – Smart Agricultural Production Engine** is developed using **Python (Flask)** and a **Random Forest Machine Learning model**. The application follows a modular project structure with separate folders for templates, static files, datasets, and the trained model. Meaningful variable names, proper code organization, and input validation improve readability and maintainability. The application successfully predicts the most suitable crop based on soil nutrients and environmental parameters. The source code is maintained in a GitHub repository for version control and future enhancements.